QUANTUMFLOOR

Computational Research Report

Job ID: 81d812ad

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'label'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	80
Unique Rows:	80
Duplicate Rows:	0
Total Columns:	5
Missing Values:	1.25%
Leakage Risk Detected:	0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	5
Quantum Embedding Dimension:	2
Qubit Budget:	2
Classical Training Size:	64
Quantum Subset Size:	64

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.5703	0.4615	0.1250	
logreg	Classical	0.5469	0.5714	0.0000	
svm_rbf	Classical	0.4844	0.6000	0.1250	
random_forest	Classical	0.4531	0.4706	0.0000	
qsvm	Quantum	0.3125	0.5556	0.1250	
pqk	Quantum	0.4375	0.5333	0.1250	
vqc	Quantum	0.6719	0.7273	0.3750	

7. Comparison & Explainability

Best Classical AUROC: 0.5469

Best Quantum AUROC: 0.6719

Delta: 0.1250

Explainability Status: available

PC2: 0.1502

PC1: 0.1166

8. Final Verdict

Best Overall Model: vqc

Quantum Advantage: Yes